

Chapter - 11.2.

Application of trees.

① Binary Search Tree: (BST)

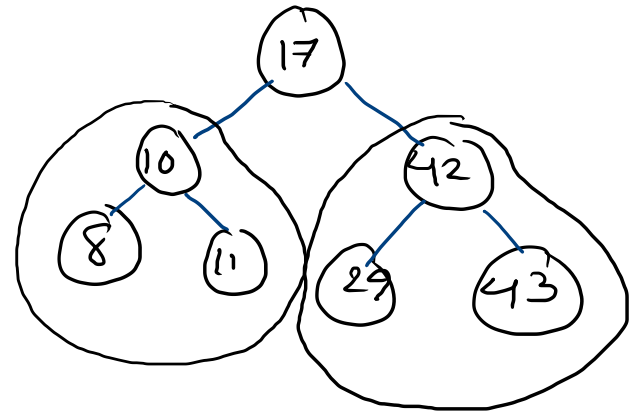
- $\text{value}_{\text{child}} < \text{parent} \rightarrow \text{left node}$
- $\text{value}_{\text{child}} > \text{parent} \rightarrow \text{right node}$

Construct BST with 17, 10, 42, 11, 8, 29, 43.

② Decision tree $\rightarrow$ m-ary tree.

$\rightarrow$ rooted tree

$\rightarrow$ internal vertices represent a decision



Subtrees.

⑩ Prefix codes.

- Encode 26 letters of EA using bit string
 - $2^5 = 32 > 26 \rightarrow |\text{bitstring}| = 5$
 - Can we encode efficiently
- less memory?
less Tx time?

⑩ Consider encoding using varying length bit-strings.

- start and end
- represent frequent letters with shorter bit strings.

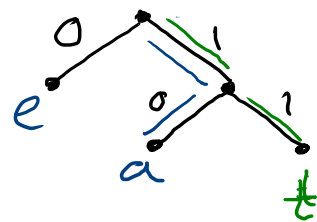
$\begin{matrix} e \rightarrow 0 \\ a \rightarrow 1 \\ t \rightarrow 01 \end{matrix} \rightarrow 0101 \rightarrow \begin{cases} tt \\ eat \\ tea \\ eaea \end{cases} \rightarrow \text{ambiguity}$

- to remove ambiguity $\rightarrow$ bit string for a letter never occurs as the first part of another letter. $\rightarrow$ Prefix codes

$\begin{matrix} e \rightarrow 0 \\ a \rightarrow 10 \\ t \rightarrow 11 \end{matrix} \rightarrow 0101 \rightarrow eat$

- Encode using rooted binary tree

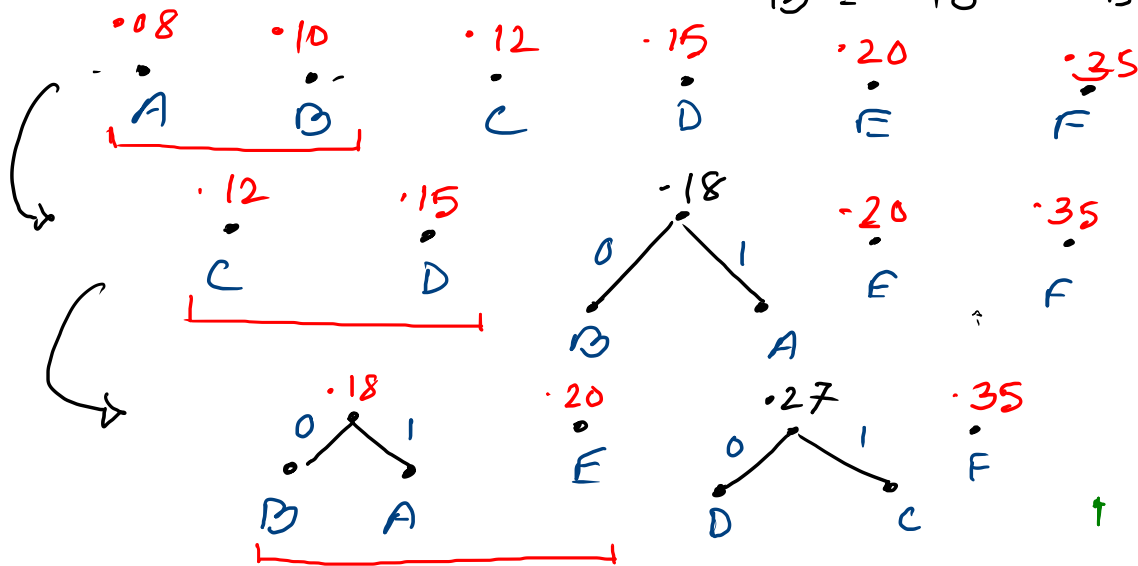
- 0 $\rightarrow$ left edge
- 1 $\rightarrow$ right edge
- chars $\rightarrow$ leaf nodes.



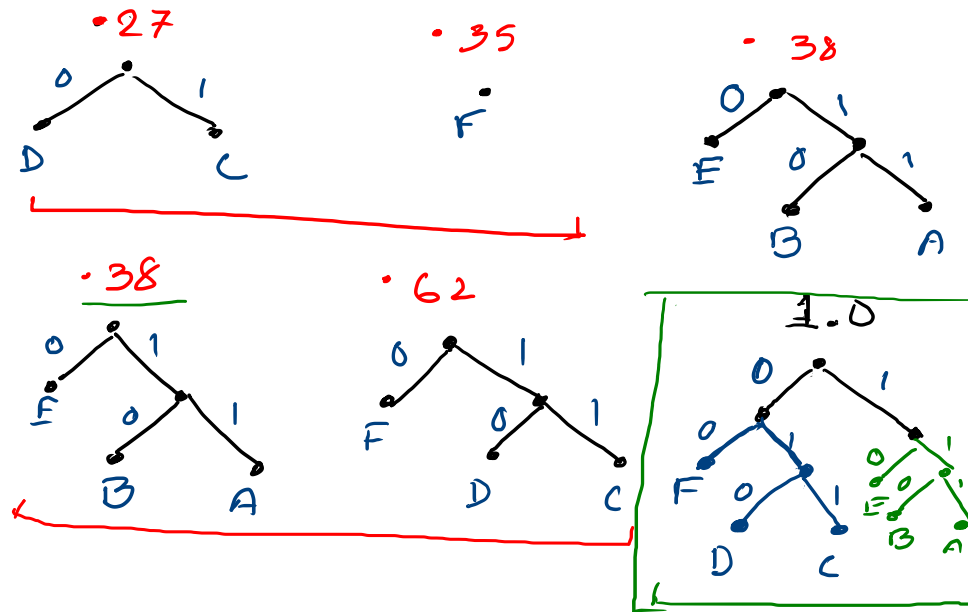
③ Huffman coding

- INPUT: Frequency/Probabilities of symbols in a string
- OUTPUT: A prefix code with the fewest possible bits
- GOAL: Given the frequencies, generate the rooted binary tree.

- Generate HC for $A = .08$ $C = .12$ $E = .20$
 $B = .10$ $D = .15$ $F = .35$



$A \rightarrow 111$ $C \rightarrow 011$ $E \rightarrow 10$
 $B \rightarrow 110$ $D \rightarrow 010$ $F \rightarrow 00$



* Game trees.

- Vertices → current state of game
- Edges → legal moves.
- Leaves → final outcome of the game.
- root → starting point.
- $\begin{cases} \text{vertex in even level} \rightarrow 1^{\text{st}} \text{ player} \\ n \quad \quad \quad n \quad \text{odd} \quad \rightarrow 2^{\text{nd}} \text{ player} \end{cases}$

* Game of Nim

* Tic-Tac-Toe